

SK-Adapter++: Enhancing Skeleton-Guided Native 3D Generation via Modality Compatible SK-Adapter

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CS-503 Project Proposal

Abstract—Recent skeleton-guided native 3D generative models such as SK-Adapter achieve precise structural control via text-and-skeleton conditioning, but their generation quality is fundamentally limited by the text-to-3D paradigm, which often produces assets with blurry textures and geometric artifacts. On the other hand, image-conditioned 3D generation offers much higher visual fidelity, yet obtaining a perfectly aligned image-skeleton pair as input is impractical in real-world use cases. We propose SK-Adapter++, a two-stage iterative refinement pipeline that combines the strengths of both paradigms. In the first stage, we use the original text-conditioned SK-Adapter to generate a coarse 3D asset that faithfully follows the skeleton. We then render this asset from a suitable viewpoint and apply an off-the-shelf image editing model (e.g., SeedREAM) to refine the rendered view, producing a high-quality reference image. In the second stage, we feed this refined image together with the skeleton into a newly trained *image-conditioned* SK-Adapter to produce the final 3D asset with both high visual fidelity and precise structural alignment. The core technical contribution is the training of an image-conditioned variant of SK-Adapter that accepts image features (instead of or in addition to text features) while retaining skeleton-based structural control.

I. INTRODUCTION

A. What is the problem you want to solve?

The creation of high-quality, structurally controllable 3D assets is a key challenge for applications in gaming, animation, and mixed reality. Recent native 3D generative models [1] can synthesize 3D assets in seconds, but they offer limited structural controllability—text prompts and image prompts cannot prescribe precise 3D articulations such as “bending the knee 60 degrees” or specifying atypical anatomical topologies.

SK-Adapter [2] addresses this gap by injecting 3D skeleton conditions into a frozen Trellis backbone via lightweight cross-attention adapters, enabling skeleton-guided text-to-3D generation with precise structural alignment. However, a critical limitation remains: **the generation quality is bounded by the text-to-3D paradigm**. Text-conditioned 3D generation often produces assets with blurry textures, loss of fine detail, and inconsistent appearance, because text descriptions inherently under-specify the visual appearance of the target object.

A natural solution is to condition generation on images instead, as image-to-3D methods typically produce significantly higher-fidelity outputs. However, in the skeleton-

guided setting, this introduces a new problem: **in practice, users do not have access to a perfectly aligned image-skeleton pair**. Users typically start with a concept (text) and a desired pose/structure (skeleton), not a pre-existing reference image that already matches the skeleton.

This creates a fundamental tension: **Text + Skeleton** $\rightarrow$ **3D** is structurally faithful but limited in visual quality while **Image + Skeleton** $\rightarrow$ **3D** has High visual fidelity but impractical due to the difficulty of obtaining a compatible image-skeleton pair.

B. Why is it important that this problem be solved?

Resolving this quality–controllability trade-off would have significant impact:

- **Production-ready assets:** Current skeleton-guided text-to-3D results still cannot achieve perfect 3D generation, which makes it limited in 3D production for industry. Higher-fidelity generation could drastically reduce this gap.
- **Enabling iterative creative workflows:** Artists typically iterate—generating a rough draft, refining it visually, and regenerating. Our proposed pipeline directly encodes this workflow into the generation process.
- **Bridging two paradigms:** Demonstrating that text-to-3D and image-to-3D can be composed in a single skeleton-conditioned pipeline opens the door to future multi-modal refinement strategies in 3D content creation.
- **Advancing skeleton-guided 3D generation:** Training an image-conditioned SK-Adapter extends the original framework to a new modality, broadening its applicability and establishing baselines for image+skeleton-conditioned 3D generation.

II. METHOD AND DELIVERIES

A. How do you solve the problem?

We propose **SK-Adapter++**, an enhanced pipeline built on top of SK-Adapter. The key idea is to *bootstrap* a high-quality reference image from the text-conditioned generation, refine it with a 2D image editing model, and then use a newly trained image-conditioned SK-Adapter to produce the final 3D asset. The pipeline consists of three stages:

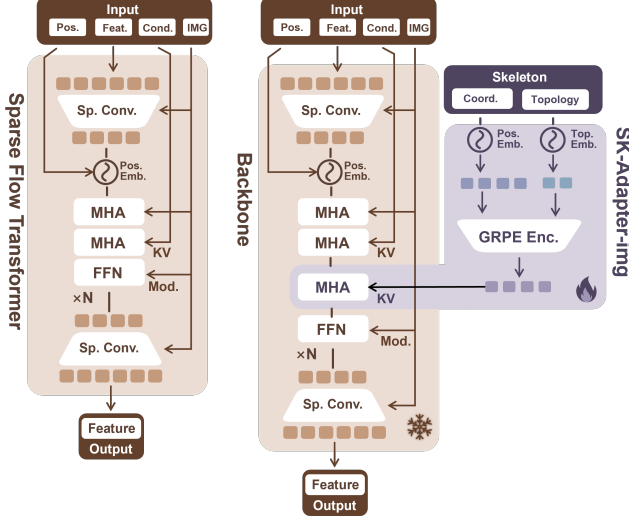


Figure 1. Overview of the SK-Adapter-img. **Stage 1:** Text + Skeleton $\rightarrow$ coarse 3D via the original SK-Adapter. **Stage 2:** Render $\rightarrow$ 2D image editing/refinement. **Stage 3:** Refined image + Skeleton $\rightarrow$ final 3D via the image-conditioned SK-Adapter.

1) *Stage 1: Coarse Generation via Text-Conditioned SK-Adapter:* Given a text prompt $\mathcal{T}$ and a 3D skeleton $\mathcal{S} = \{J, G\}$, we use the pre-trained SK-Adapter to generate an initial 3D asset $\mathcal{X}_{\text{coarse}}$:

$$\mathcal{X}_{\text{coarse}} = \mathcal{F}_{\text{text}}(\mathcal{T}, \mathcal{S}). \quad (1)$$

This asset is structurally aligned with the skeleton but may suffer from the limited visual quality typical of text-to-3D generation.

2) *Stage 2: 2D Image Refinement:* We render $\mathcal{X}_{\text{coarse}}$ from a carefully selected viewpoint (e.g., the frontal canonical view or the view that maximizes visible surface area) to obtain a reference image I_{coarse} . We then apply an off-the-shelf image editing/generation model $\mathcal{E}_{2D}$ (e.g., SeedREAM [3], or other controllable image generation models) to refine this image:

$$I_{\text{refined}} = \mathcal{E}_{2D}(I_{\text{coarse}}, \mathcal{T}), \quad (2)$$

where $\mathcal{T}$ or additional style prompts guide the refinement. The 2D editing model enhances texture quality, adds fine details, and corrects visual artifacts while broadly preserving the pose and structure from the rendered view. This step leverages the powerful priors of state-of-the-art 2D generative models, which are substantially more mature than their 3D counterparts.

3) *Stage 3: Final Generation via Image-Conditioned SK-Adapter:* The refined image I_{refined} and the original skeleton $\mathcal{S}$ are fed into our newly trained *image-conditioned SK-Adapter* to produce the final high-quality 3D asset:

$$\mathcal{X}_{\text{final}} = \mathcal{F}_{\text{img}}(I_{\text{refined}}, \mathcal{S}). \quad (3)$$

Training the Image-Conditioned SK-Adapter. The core technical contribution is training an image-conditioned vari-

ant of SK-Adapter. The Trellis backbone [1] already supports image conditioning via DINOv2 [4] features (as opposed to CLIP text features used in the text-conditioned mode). We train a new set of SK-Adapter modules (skeleton encoder + cross-attention layers) on top of the *image-conditioned* Trellis backbone, following the same PEFT strategy as the original SK-Adapter:

- The Trellis backbone is **frozen** in image-conditioning mode.
- A GRPE-based skeleton encoder processes the 3D skeleton into topology-aware tokens.
- Trainable cross-attention layers inject these skeleton tokens into the frozen backbone.
- Zero-initialized linear projections ensure non-invasive integration at training onset.

The training objective remains the Latent Flow Matching loss:

$$\mathcal{L}_{FM} = E_{t, \mathbf{z}_0, \mathbf{z}_t} \|v_{\theta}(\mathbf{z}_t, t, \mathbf{c}_{\text{img}}, \mathbf{f}_{\text{skel}}) - \mathbf{u}_t(\mathbf{z}_0)\|^2, \quad (4)$$

where $\mathbf{c}_{\text{img}}$ are the DINOv2 image features (replacing $\mathbf{c}_{\text{text}}$). Training data is constructed from the Objaverse-TMS dataset by rendering reference views of each 3D asset.

B. How will you validate your solution?

We will evaluate on two axes: **structural alignment** and **visual fidelity**, following the evaluation protocol of the original SK-Adapter. (**Note that the evaluation framework of SK-Adapter is very complicated for a course project. So the plan is provisional. We might adopt a more simplified plan.**)

Metrics:

- **ReRigging Score** ($\downarrow$): Chamfer Distance between a re-extracted skeleton from the generated mesh (via Any-mate [?]) and the input skeleton. Measures structural faithfulness.
- **CLIP Score** ($\uparrow$): Cross-modal similarity between multi-view renders and the text prompt. Measures semantic consistency.
- **PickScore** ($\uparrow$): Human-aligned perceptual quality score for generated views.
- **KD-DINO** ($\downarrow$): Kernel Distance computed with DINOv2 features between generated and ground-truth multi-view distributions. Measures distributional visual fidelity.

Baselines:

- SK-Adapter (text + skeleton only, no refinement) [2]
- SKDream (2D-projection-based skeleton conditioning) [5]
- Trellis image-to-3D (image only, no skeleton conditioning) [1]

III. RELATED WORK

Native 3D generative models. The field has rapidly advanced from optimization-based methods [6], [7] to feed-forward native 3D generation [8], [9], [1]. Trellis [1] represents the state of the art, generating 3D assets from text or images via sparse-structure flow transformers operating on Structured Latent (SLaT) representations. Trellis supports both text and image conditioning (using CLIP and DINOv2 features, respectively), but provides no mechanism for explicit structural control.

Skeleton-guided 3D generation. In 2D, ControlNet [10] and T2I-Adapter [11] have demonstrated that skeletal pose signals can precisely control image layout. Extending this to 3D, SKDream [5] projects 3D skeletons into 2D maps to condition multi-view diffusion, but this “2D lifting” strategy suffers from spatial ambiguity and multi-view inconsistency. SK-Adapter [2] resolves this by operating entirely in the native 3D domain: it encodes the 3D skeleton as sparse topology-aware tokens via GRPE and injects them into a frozen Trellis backbone via cross-attention. However, SK-Adapter is limited to text conditioning, inheriting the visual quality ceiling of text-to-3D generation.

IV. DISCUSSION

A. Implications and Contribution

We expect SK-Adapter++ to achieve:

- **Improved visual fidelity** (higher PickScore, lower KD-DINO) compared to the text-only SK-Adapter, while maintaining comparable or better structural alignment (similar or lower ReRigging Score).
- **A practical generation workflow:** the pipeline requires only text + skeleton as user input (no manual image preparation), making it directly usable.
- **An image-conditioned SK-Adapter model:** this is a standalone contribution—an image+skeleton-conditioned 3D generative model enables new use cases (e.g., reconstructing a 3D character from a reference image in a specific pose).

B. Shortcomings

The 2D editing model may alter the pose/structure, causing misalignment with the skeleton in Stage 3.

REFERENCES

- [1] J. Xiang, Z. Lv, S. Xu, Y. Deng, R. Wang, B. Zhang, D. Chen, X. Tong, and J. Yang, “Structured 3d latents for scalable and versatile 3d generation,” 2025. [Online]. Available: <https://arxiv.org/abs/2412.01506>
- [2] A. Wang, Y. Ao, S. Wu, and C.-K. Tang, “Sk-adapter: Skeleton-based structural control for native 3d generation,” 2026. [Online]. Available: <https://arxiv.org/abs/2603.14152>
- [3] T. Seedream, :, Y. Chen, Y. Gao, L. Gong, M. Guo, Q. Guo, Z. Guo, X. Hou, W. Huang, Y. Huang, X. Jian, H. Kuang, Z. Lai, F. Li, L. Li, X. Lian, C. Liao, L. Liu, W. Liu, Y. Lu, Z. Luo, T. Ou, G. Shi, Y. Shi, S. Sun, Y. Tian, Z. Tian, P. Wang, R. Wang, X. Wang, Y. Wang, G. Wu, J. Wu, W. Wu, Y. Wu, X. Xia, X. Xiao, S. Xu, X. Yan, C. Yang, J. Yang, Z. Zhai, C. Zhang, H. Zhang, Q. Zhang, X. Zhang, Y. Zhang, S. Zhao, W. Zhao, and W. Zhu, “Seedream 4.0: Toward next-generation multimodal image generation,” 2025. [Online]. Available: <https://arxiv.org/abs/2509.20427>
- [4] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, M. Assran, N. Ballas, W. Galuba, R. Howes, P.-Y. Huang, S.-W. Li, I. Misra, M. Rabbat, V. Sharma, G. Synnaeve, H. Xu, H. Jegou, J. Mairal, P. Labatut, A. Joulin, and P. Bojanowski, “Dinov2: Learning robust visual features without supervision,” 2024. [Online]. Available: <https://arxiv.org/abs/2304.07193>
- [5] Y. Xu, Z. Yang, and Y. Yang, “Skdream: Controllable multi-view and 3d generation with arbitrary skeletons,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2025, pp. 314–325.
- [6] B. Poole, A. Jain, J. T. Barron, and B. Mildenhall, “Dreamfusion: Text-to-3d using 2d diffusion,” 2022. [Online]. Available: <https://arxiv.org/abs/2209.14988>
- [7] Z. Wang, C. Lu, Y. Wang, F. Bao, C. Li, H. Su, and J. Zhu, “Prolificdreamer: High-fidelity and diverse text-to-3d generation with variational score distillation,” 2023. [Online]. Available: <https://arxiv.org/abs/2305.16213>
- [8] Y. Hong, K. Zhang, J. Gu, S. Bi, Y. Zhou, D. Liu, F. Liu, K. Sunkavalli, T. Bui, and H. Tan, “Lrm: Large reconstruction model for single image to 3d,” 2024. [Online]. Available: <https://arxiv.org/abs/2311.04400>
- [9] J. Xu, W. Cheng, Y. Gao, X. Wang, S. Gao, and Y. Shan, “Instantmesh: Efficient 3d mesh generation from a single image with sparse-view large reconstruction models,” 2024. [Online]. Available: <https://arxiv.org/abs/2404.07191>
- [10] L. Zhang, A. Rao, and M. Agrawala, “Adding conditional control to text-to-image diffusion models,” 2023. [Online]. Available: <https://arxiv.org/abs/2302.05543>
- [11] C. Mou, X. Wang, L. Xie, Y. Wu, J. Zhang, Z. Qi, Y. Shan, and X. Qie, “T2i-adapter: Learning adapters to dig out more controllable ability for text-to-image diffusion models,” 2023. [Online]. Available: <https://arxiv.org/abs/2302.08453>